

Creation of a Suburban Rail Service in Khartoum



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Cette étude porte sur la création d'une desserte suburbaine à Khartoum au Soudan. La concentration des points d'intérêts au centre-ville et la décentralisation de la population croissante dans les banlieues, rendent nécessaire une nouvelle solution de transport. Ce rapport présente un diagnostic complet de la situation, suivi d'une analyse multicritère permettant de choisir une solution. Cette solution est ensuite dimensionnée, et sa mise en œuvre est détaillée, en termes d'infrastructures, matériel roulant et exploitation. Finalement, un bilan socio-économique de la solution est réalisé, incluant une analyse de risque détaillée. Le résultat est une réhabilitation complète de la ligne ferroviaire existante. La nouvelle desserte proposera des trains fréquents, avec des matériels roulants hybrides neufs. Tous les équipements seront prévus pour permettre une augmentation progressive des flux de passagers, anticipant ainsi l'adoption croissante par la population de la solution ferroviaire..

Mots-clés : analyse multicritères mobilité, desserte ferroviaire, évaluation socio-économique, matériel roulant hybride, report modal, Khartoum

This study focuses on the creation of a suburban service in Khartoum, Sudan. The concentration of points of interest in the city center and the decentralization of the growing population in the suburbs make a new transportation solution necessary. This report presents a comprehensive diagnosis of the situation, followed by a multi-criteria analysis to choose a solution. This solution is then sized, and its implementation is detailed in terms of infrastructure, rolling stock, and operation. Finally, a socio-economic assessment of the solution is carried out, including a detailed risk analysis. The result is a complete rehabilitation of the existing railway line. The new service will offer frequent trains with new hybrid rolling stock. All equipment will be designed to allow a gradual increase in passenger flows, thus anticipating the population's growing adoption of the new rail solution.

Keywords: hybrid rolling stock, mobility, modal report, multi-criteria analysis, rail service, socio-economic analysis, Khartoum

1. Context of the study

1.1. General Context

Home to almost 48 million inhabitants, Sudan is, at the time of this study, in the midst of a violent civil war. This study places itself in the context of the end of the war and considers the development of a new railway service in and around Khartoum as part as the economic redevelopment of the country.

Even though the Sudan has had an extensive railway network since the end of the 19th century, this network is now in a derelict state. At the country level, improving the railway system is a strategic lever to help the economic situation. Indeed, Sudan has large reserves of oil and ore, which a good transportation system will help exploit and connect with the country harbors.

1.2. Mobility Needs in Khartoum

Khartoum is the capital city of Sudan, and also its most densely populated region. As its population keeps growing, it also expands in the suburban areas around the city center, in a radius reaching over 20km. Most of the service and work areas being in the city center, this generates high levels of congestion on the streets during rush hour.

The main current modes of transportation are the minibus, the personal car and people traveling by foot. Minibus services are not organized, they don't follow specific routes and stop on demand. This also contributes to traffic issues in the capital.

A railway line already exists in the region, traversing the city from South to North and stopping at three stations in the city. However, it is not used anymore on most of its length, due to the lack of maintenance on the infrastructure.

2. Choice and Scaling of the Solution

2.1. Multi-criteria Analysis

A multi-criteria analysis was carried out to select a transport solution and a route. The choice of transport solution was based six main criteria, all centered around sustainable mobility.

2.1.1. Choice of transport solution

Four solutions were compared to select the one best suited to Khartoum's needs:

- Shared Bus Rapid Transit (BRT) line;
- Khartoum Metro Line;
- Tramway line in Khartoum;
- Regional Express Railway Line in Khartoum.

Given the country's financial limitations, the best transport capacity/investment/robustness ratio was a priority.

Following this analysis, the following conclusions were reached:

- The rail system offers the greatest transport capacity;
- The pre-existence of the infrastructure will significantly reduce the cost of acquisition and the time required to set up the target transport system;
- The positioning of the existing stations ensures spatial coverage of the service points and is consistent with a possible extension.

2.1.2. Choice of route

As the rail solution better meets Khartoum's mobility needs, three potential solutions for getting to and/or from the city center have been considered:

- Route 1: Rail link from the center of Khartoum to the west.
- Route 2: Rail link from the center of Khartoum to the north.
- Route 3: Route for a rail link between South and Central Khartoum

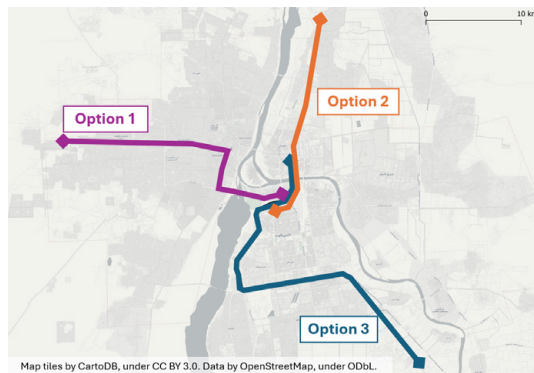


Figure 1. Three routes studied. © Personal creation with map tiles by CartoDB

The descriptions and main constraints associated with each of these routes are set out in the table below:

	Route 1	Route 2	Route 3
Description	Center of Khartoum to the West	Center of Khartoum to North	South and Center of Khartoum
Main Constraints	-No existing railway line -Taking the train over the Nile branch & building a bridge	- Low population density covered - Along a main road & Illegal level crossings slightly below 80	-High population density covered -Coverage of several business areas -86 level crossings -Easy access to Khartoum Airport

Based on this analysis, the solution chosen is a railway service along route 3, providing access to the city center and the south of the city. The solution will be called "KER: Khartoum Express Railways".

2.2. Scale of the Main Variables

2.2.1. PPHPD

The main variable that will help determine the size of the service is the number of passengers per hour per direction, or PPHPD.

A PPHPD ranging from 3750 to 10000 can be estimated based on the number of trips between the suburbs and the city center per day (2 million) and the hypothesis made on the expected modal shift.

Of this range, it was decided to consider the lower bound to avoid over-sizing the solution. This is taking into account a progressive modal shift. However, the fixed infrastructure and system will be sized to allow for PPHPD growth in the future without needing any major change.

2.2.2. Interval

The proposed initial interval is 15 minutes. It is short enough to guarantee a frequent service, and corresponds to an easy-to-remember schedule.

2.2.3. Max speed

The track infrastructure will be designed to allow a maximum speed of 80km/h. Even though initially the operating speeds will be slower, this speed of 80km/h will allow an increased PPHPD in the future and doesn't require restraining the rolling stock below its full capacity.

2.2.4. Number of tracks

The existing line has only one single track. As part of this study, it was decided to upgrade the line to a double track. This will ease the operations and hopefully lower the number of incidents and delays. The operation will also be more robust, as an issue on one track should not stop the service in the other direction.

3. Implementation of the Solution

3.1. Systems and Infrastructure

3.1.1. Tracks

The line will keep the current Cape gauge of 1.067m, as this line is part of the national network. The tracks will be laid on sleepers and ballast, as a concrete slab solution would be too expensive and challenging to maintain. Ballast will also help with draining water, as the region is prone to flooding.

The minimum bending radius is calculated at 472 m, to allow a maximum speed of 80 km/h.

3.1.2. Stations

On top of the three stations already existing, it is proposed to create eight new stops along the line. One major stop at the southern terminus will also include a new rolling stock maintenance facility.

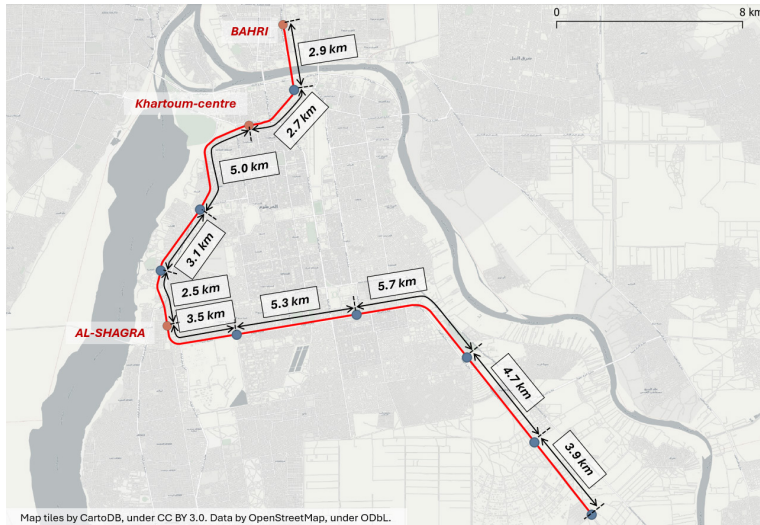


Figure 2. Station locations © Personal creation with map tiles by Carto DB

3.1.3. Bridges and Crossings

As the line is currently unused, it is crossed by streets in over 86 places.

As part of the line renewal, it is proposed to drastically reduce the number of crossings, so that the trains can operate smoothly. Four new bridges will be constructed on the main street crossings, and only twenty level crossings will be kept.

3.1.4. Signaling Systems

The choice of signaling system falls on the automatic three-way light block, which is a relatively simple and reliable solution.

Signals will be placed at least every 1500 m, which, at 80km/h, allows a minimum interval of 149s, which is well below the initial 15 minutes considered.

3.1.5. Maintenance

A large maintenance facility already exists next to Khartoum central station. It will probably need to be renovated, but it is a large space available to store trains, with more than 20 side tracks available. As part of this new suburban service project, a new 1Ha maintenance facility will also be built at the south terminus station.

3.2. Rolling Stock

Given the poor electrical power production in SUDAN, and using a benchmark on projects in similar environments, like the TER in Dakar, we are able to choose the appropriate kind of equipment and its size.

The rolling stock that best meets our needs is called "Régional Hybrid Train series B 83500", from the Coradia family made by the French manufacturer Alstom. Initially a dual-mode train (catenary / diesel), modifications which were made to the train consist of replacing two of the four diesel engines (4*300kW) with lithium-ion batteries which allow a range up to 30 km thanks to batteries and 1,000 km with diesel and battery combined. These batteries can recycle 90% of the braking energy.

The self-propelled train chosen is a low-floor peri-urban model with two doors per side on each vehicle, and offering carrying total capacity of 470 passengers per railcar.

To cover our peak hour transport needs, i.e. 938 passengers, the train will be made up of 2 trainsets with a total length of 72m x 2 = 144m which will be parked on 200m long station platforms.

3.3. Operations and Maintenance

The existing railway line traversing Khartoum from north to south accommodates increasing travel demand, driven by rapid urbanization, particularly in the suburban areas. The previous gradual abandonment of this railway line makes modal shift complex. Therefore, this shift cannot occur without genuine attractiveness, both in terms of fare pricing and the service offered.

3.3.1. Passenger Service

To best meet the mobility needs of Khartoum's residents, a daily service from Sunday to Thursday between 05:00 and 22:00 will be provided, with three different headways based on the expected passenger traffic, as follows:

Table 1. Headway based on time slot

Headway (min)	Time Slots	
15	from 6:30 am to 12:00 pm	from 2:00 pm to 5:00 pm
20	from 12:00 pm to 2:00 pm	from 5:00 pm to 8:00 pm
30	from 5:00 am to 6:30 am	from 8:00 pm to 10:00 pm

To ensure better take-up of the service, an omnibus service is preferred. Regular scheduling will be an advantage compared to the long waits observed at bus stations.

3.3.2. Operation

The travel time between the two termini is 57 minutes during peak hours. The total distance is 39.5 km, and the chosen commercial speed is 41.5 km/h. We estimate the turnaround time, including buffer time, to be 25 minutes per terminus. Therefore, a complete loop will take 2 hours and 20 minutes.

To facilitate the inevitable transfer breaks, intermodal hubs must be organized. The current locations of mini-bus clusters partly dictated the positioning of the railway stations.

- Rolling Stock

Based on the interval during peak hours and the total travel time, the following formula calculates the number of trains needed during peak hours, resulting in a requirement of 9.27 trains.

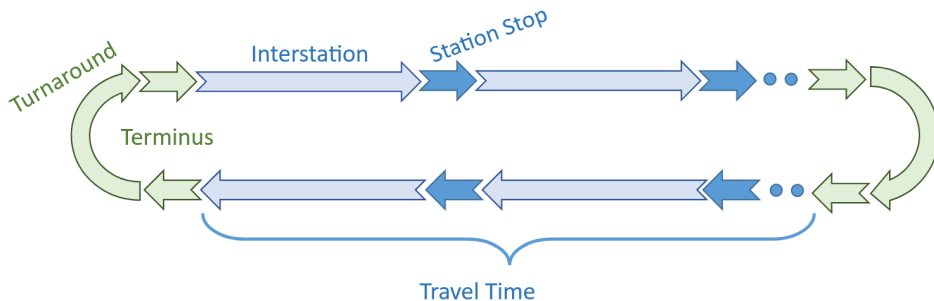


Figure 3. Schematic representation of variables for calculating the number of trains © Myriam Essafi

Applying a 25% maintenance ratio and provisioning an additional vehicle to address operational incidents, the total number of trains required is 13.

- Maintenance

Maintenance will primarily be conducted at night, but a daytime team will be necessary for more extensive maintenance tasks and to address daily operational contingencies.

- Timetable

Based on the distances between stations and the acceleration and braking capabilities of the rolling stock, we achieve an average inter-station speed of 60 km/h. From this, we can derive the following space-time graph for peak hours:

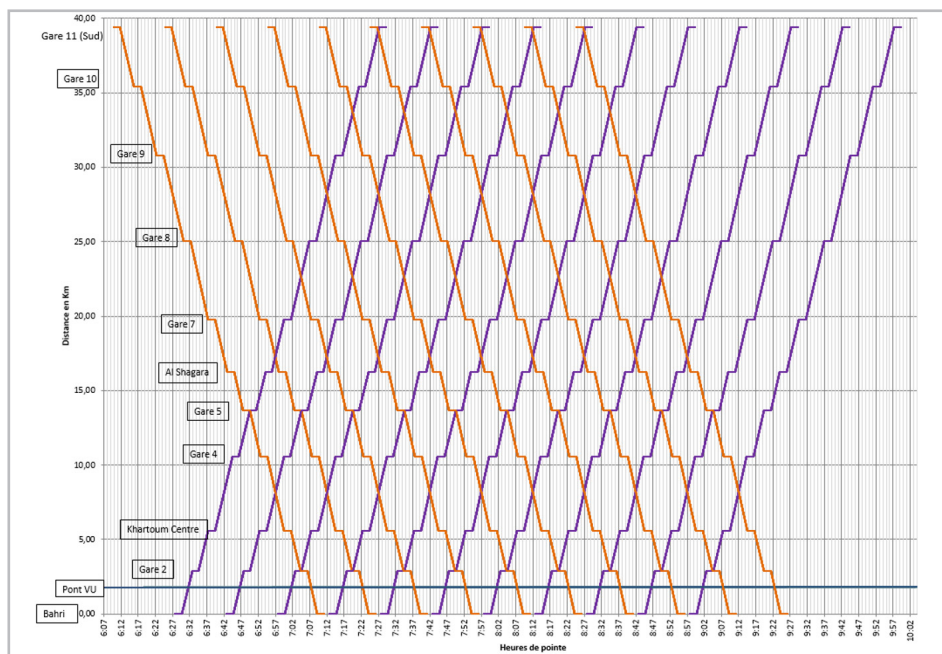


Figure 4. Timetable during peak hours © Myriam Essafi

The main challenge is ensuring that no two trains are scheduled simultaneously on the Blue Nile Bridge section, which consists of a single track. This area is represented on the space-time graph by the line labeled "Pont VU," where it is evident that no two trains ever cross paths.

- Rolling Stock Rotation

The rotation schedules for the rolling stock have been designed to cover all trips in the timetable. The trains will complete their rotations at the station where they need to start the following day. Therefore, no empty trips will be necessary. The driver rotations will also be optimized for passenger service.

3.3.3. Stations

The main stations are through stations. Their organization is simplified as they are equipped with double tracks, allowing dedicated directions of travel under normal conditions. The two terminal stations already have the infrastructure to store trains and enough space for expanding a maintenance center. They will be considered terminal stations where trains will turn around.

3.3.4. Fares and ticketing

The choice of ticket price will be crucial for the service's attractiveness. It will be calculated considering the current fares of other public transportation modes. A flat fare will be preferred across the entire line. To attract frequent travelers with daily commuting needs, weekly, monthly, and even annual subscriptions should be offered at suitable rates. Similarly, student fares should be part of the offering.

The main motivation for maintaining a low fare is that it is essential for encouraging modal shift. It could be beneficial to offer a combined ticket that allows a journey by train and mini bus. However, this reduced revenue will inevitably impact the operational balance.

3.4. Solution deployment schedule

The estimated schedule from pre-studies phase to commissioning is 5 years, including selection of foreign stakeholders which should be the major challenge for this project in addition to funding.

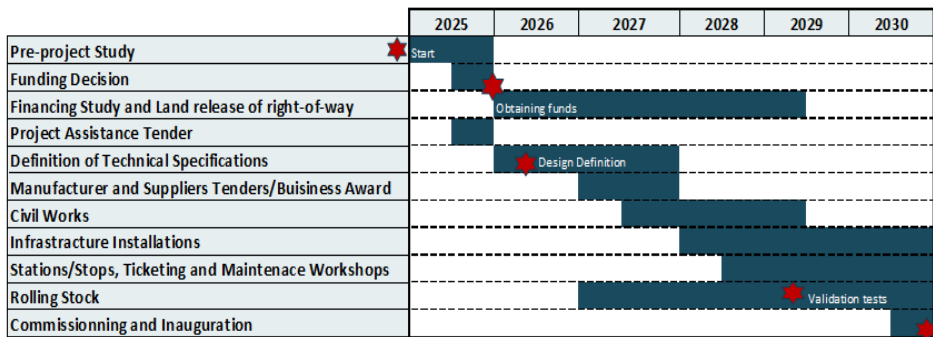


Figure 5. Project Planning

4. Socio-Economic Evaluation

4.1. Financial Analysis

Following the breakdown for all associated costs, an investment of €1.6 bn is the overall budget covering the project steps from studies to the launch of the commercial service.

In order to maintain operational services, an allocated package of €89 M is necessary as a yearly expense.

Total CAPEX costs are around €40 M/km and €2.2 M/km/year.

Table 2. Total Financial Package

	Cost (EUR)	Cost per km (EUR)
CAPEX	€1, 612,254,767	€40,306,369
OPEX/year	€88,740,000	€2,218,500

The total budget is split as follows for each cost center:

Table 3. Budget Breakdown

Cost Center	Cost (€M)	%
Infrastructure	603	35.46%
Civil Works	506	29.73%
Rolling Stock	172	10.12%
Stations/Stops, Ticketing	150	8.82%
Construction Supervision Services	77	4.53%
Risk provisions	71	4.12%
Operation	56	3.29%
Project Assistance	43	2.47%
CSR impact for modal report/ Environmental File	21	1.24%
Services	3	0.20%
Land and release of right-of-way	0.21	0.01%
Total CAPEX/OPEX	€1.8 bn	100%

Based on the hypothesis of €200 M funding provided by AFD for this project, there will be a €1.4 bn shortfall in achieving the total amount; therefore, another financial backer should be involved.

The World Bank would be the best option considering the amount of funding needed, the socio-political situation of Sudan, and also the collaboration with AFD.

Table 4. Financial Funds

AFD Funding	€200,000,000	12%
Financial backer (to be defined)	€1,412,254,767	88%

For the economic viability of the project, the results of net present value are significantly negative considering an amortization over 50 years.

Table 5. Project economic viability

Initial investment (€)	€1,612,254,767
Duration (years)	50
Actualisation rate/ year	45%
Public Subvention/ year	20%
Wear rate / year	3%
Annual incomes	€48,960,000
Annual costs	€88,740,000
VAN	-€781 425 875

However, numerous externalities are identified as having an impact on the propulsion of the development of socio-economic conditions of Khartoum's population, especially since the motorization rate in Sudan is around 4%, and that the more vulnerable citizens depend on public transport to access essential services and workplaces.

4.2. Risk analysis

The implementation of any new project involves risks. It is essential to identify, evaluate, and propose means of managing these risks and implementing and prevention actions. The starting hypothesis of this project is the end of the current war in the country. This topic will not be included in the risks considered.

4.3. Risk management plan

After identifying up to 21 risks and classifying them according to their criticality (C), calculated using the formula $C = F \times G \times D$, a full risk management plan was established.

The three most critical risks retained, and the associated action plans are as follows:

Risk	C	Actions Planned
Insufficient funding	25	-Improve monitoring of the management of funds -Involve public and/or private actors in financing -Diversify financial partners
Personal accidents	25	-Increase awareness-raising actions among local populations -Install fences along the line, over its entire length or on specific sections
Collision with obstacles	20	-Awareness-raising actions among local populations -Passage of the track into a protected area via fences

Conclusion

The Present pre-opportunity project proposes the creation of a suburban rail service in Khartoum to improve urban mobility and respond to increasing urbanization.

The "KER - Khartoum Express Railways" project aims to reinstate 39 km of railway tracks, create 11 stations and offer 4 trains per hour in each direction with Régiolis Hybrid Diesel/battery type rolling stock. The estimated cost is in the order of 40 million euros per km, including rehabilitation, construction of new stations and acquisition of rolling stock.

The expected benefits include a reduction in road congestion, improved accessibility and a positive impact on the local economy and the environment. Additionally, the rail solution helps to significantly reduce the city's carbon footprint, aligning the project with environmental sustainability goals.

To ensure the success of this initiative, it will be crucial to maintain a high level of coordination between government actors, international donors and local partners. Collaboration with existing public transport companies and minibus unions and a strategic approach to intermodality will be essential to ensure a smooth transition to the new transport system.